Data Dictionary

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This document defines the variables contained in the four CSV files deposited in the repository (https://github.com/miltheo/SCOPE-MOVE) in folder /analysis/inputs. Definitions and extraction rules follow the project extraction manual (https://osf.io/bk7yr/files/yk8a3), with additional derived variables used for harmonised reporting and meta-analysis in the accompanying study (future citation).

Missing data codes

Code	Meaning
n.r.	Not reported in the source study.
n.a.	Not applicable for the study/model.
None	Explicitly not done / not used (as stated by authors).
Unclear	Could not be determined after full-text review and checking cited sources.

Where these codes appear as literal strings in a column, treat them as missingness indicators rather than valid values.

Files and relational structure

1) Study_characteristics.csv: one row per included study (study-level metadata). Primary key: id.

2) Extraction_Models_Master Sheet with Validation IDs.csv: one row per classifier validation (validation-level). Primary keys: val_id, with study identifier id for joining to Study_characteristics.

3) Extraction_Energy Expenditure Models with Validation IDs.csv: one row per energy expenditure validation (validation-level). Primary keys: val_id, with study identifier id for joining to Study_characteristics.

4) Quality Assessment.csv: QUADAS-2 signalling questions and domain judgements, one row per study. Joins to Study_characteristics via Study ID (author-year).

Extraction_Models_Master Sheet with Validation IDs.csv

Rows: 210. Columns: 38.

Column	Type	Description	Allowed/Example values	Notes
val_id	integer	Unique validation identifier (validation-level row id).		Primary key for validation-level tables.
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_characteristics.id to validation-level tables.
study	text	Study ID in the format First author (year).		Corresponds to 'Study ID' in the extraction manual.
Is the model published previously?	text	Whether the evaluated model was previously published before the validation study.		
movement_behaviour	categorical	Movement behaviour domain assessed by the prediction model.	Activity intensity; Activity type; Sleep-wake	
model_type	text	High-level model type (e.g., multi-class classifier).	Binary classifier; Multi-class classifier; Multiple binary classifiers	
model_test_outcomes	text	Outcome classes or continuous outcome specification used for testing.		
model_test_classification	text	Explicit classification problem definition (e.g., sleep vs wake).		
test_criterion	text	Criterion/reference method used for validation.	DO; DO; VO; IC; PSG; VO	Common codes: DO direct observation, VO video observation, PSG polysomnography, IC indirect calorimetry, DLW doubly labelled water.
Reported?	text	Indicator that the performance metric was directly reported vs derived.		
CM?	text	Indicator that a confusion matrix with counts was available for classifiers.	%; EC; NO	% for percentage reporting, EC for event counts; NO when not reported
age_group	categorical	Age group category for the cohort used for model development/testing.	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
test_device	text	Wrist device used in the validation.		

test_model	text	Specific prediction model/method name as reported.		
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries.
wear_site	text	Wear site for the device (e.g., non-dominant wrist).		
features	text	Types of features input in the prediction model.	Time-domain; Frequency-domain; Spatial; Time-Frequency; Non-linear	
single_feature	object	Name of single feature used with unit in parenthesis.	MAD (m*g*), ENMO (m*g)*	n.a. if multiple input features
method	categorical	Method family used for grouping (classifiers table).	Threshold / Cutpoint; Heuristic Rule; Linear / GLM; Classical ML; Deep Learning; Probabilistic Temporal; Hybrid / Ensemble	
method_sub	text	More specific method label within the method family.		
train_environment	text	Environment where training data were collected.	Free-living; Hybrid; Laboratory	
train_protocol	text	Training protocol structure.	Hybrid; Structured; Unstructured	
test_environment	text	Environment where testing/validation data were collected.	Free-living; Hybrid; Laboratory	
test_protocol	text	Testing protocol structure.	Hybrid; Structured; Unstructured	
validation_type	text	Validation approach used for evaluation.	Apparent; Cross-validation; External; Hold-out	
f1_mean	numeric	Macro-averaged F1 score (percentage, 0-100).		
f1_sd	numeric	Standard deviation or uncertainty for F1 (percentage points).		
sens_mean	numeric	Sensitivity (true positive rate) (percentage, 0-100).		
sens_sd	numeric	Standard deviation or uncertainty for sensitivity (percentage points).		
spec_mean	numeric	Specificity (true negative rate) (percentage, 0-100).		
spec_sd	numeric	Standard deviation or uncertainty for specificity		

		(percentage points).		
metric_origin	categorical	How class-based metrics were averaged/derived.	macro; micro; per_class; aggregate_cm; unclear; n.a.	
sd_origin	categorical	Source of SD/SE for the metric.	cv_folds; classes; participants; cv+classes; external_test; not_reported; n.a.	
k_folds	text	Number of folds/replicates in the outer validation loop.		Stored as text in CSV; numeric when available.
n_classes	text	Number of outcome classes over which performance was computed.	10; 2; 3; 4; 5; 6; n.a.	
n_participants	text	Number of unique participants contributing to the validation/test set.		Stored as text to preserve n.a./n.r. codes.
other	text	Free-text flags and notes used during extraction and harmonisation.		
doi	text	Digital Object Identifier or persistent link for the study.		

Extraction_Energy Expenditure Models with Validation IDs.csv

Rows: 67. Columns: 32.

Column	Type	Description	Allowed/Example values	Notes
val_id	integer	Unique validation identifier (validation-level row id).		Primary key for validation-level tables.
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_characteristics.id to validation-level tables.
study	text	Study ID in the format First author (year).		Corresponds to 'Study ID' in the extraction manual.
Is the model published previously?	text	Whether the evaluated model was previously published before the validation study.	Yes; No - Hildebrand 2014	
movement_behaviour	categorical	Movement behaviour domain assessed by the prediction model.	Energy expenditure	
model_type	text	High-level model type (e.g., multi-class classifier).	Supervised regression	
model_test_outcomes	text	Outcome classes or continuous outcome specification used for testing.		
model_test_classification	text	Explicit classification problem definition (e.g., sleep vs wake).	n.a.	
test_criterion	text	Criterion/reference method used for validation.	DLW; IC; VO	Common codes: DO direct observation, VO video observation, PSG polysomnography, IC indirect calorimetry, DLW doubly labelled water.
age_group	categorical	Age group category for the cohort used for model development/testing.	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
test_device	text	Wrist device used in the validation.		
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries. Healthy is apparently healthy.
wear_site	text	Wear site for the device (e.g., non-dominant wrist).	Both wrists; ; Dominant wrist; Left wrist; Non-dominant wrist; Right wrist; n.r.	
features	text	Types of features input in the		

		prediction model.		
single_feature	object	Name of single feature used with unit in parenthesis.	MAD (m*g*), ENMO (m*g)*	n.a. if multiple input features
method	categorical	Method family used for grouping (classifiers table).	Threshold / Cutpoint; Heuristic Rule; Linear / GLM; Classical ML; Deep Learning; Probabilistic Temporal; Hybrid / Ensemble	
method_sub	text	More specific method label within the method family.		
method_group	categorical	Binary grouping of methods (energy expenditure table).	Threshold/Heuristic or Linear Models; Machine Learning or Other Models	
train_environment	text	Environment where training data were collected.	Free-living; Hybrid; Laboratory	
train_protocol	text	Training protocol structure.	Hybrid; Structured; Unstructured	
test_environment	text	Environment where testing/validation data were collected.	Free-living; Hybrid; Laboratory	
test_protocol	text	Testing protocol structure.	Hybrid; Structured; Unstructured	
validation_type	text	Validation approach used for evaluation.	Apparent; Cross-validation; External; Hold-out	
mape_mean	numeric	Mean absolute percentage error for energy expenditure estimation (percentage).		
mape_sd	numeric	Standard deviation or uncertainty for MAPE (percentage points).	12.97; 14; 15.1; 4.57; 5.1; 5.26; 7.89; 9.49; 9.52; n.r.	
rmse_met_mean	numeric	Root mean square error in METs, if reported.		
rmse_met_sd	numeric	Standard deviation or uncertainty for RMSE in METs.	0.028; 0.059; 0.06; 0.13; 0.21; 0.36; 0.67; 1.7; n.r.	
n_participants	text	Number of unique participants contributing to the validation/test set.	10; 15; 17; 24; 30; 33; 36; 51; 59; 7; 9; n.a.	Stored as text to preserve n.a./n.r. codes.
sd_origin	categorical	Source of SD/SE for the metric.	cv_folds; classes; participants; cv+classes; external_test; not_reported; n.a.	
k_folds	text	Number of folds/replicates in the outer validation loop.	10; 15; 17; 30; 59; 7; 9; n.a.	Stored as text in CSV; numeric when available.

other	text	Free-text flags and notes used during extraction and harmonisation.		

Quality Assessment.csv

Rows: 137. Columns: 18.

See https://osf.io/bk7yr/files/a8sq7 for the Quality Assessment tool or refer the original study.

Column	Type	Description	Allowed/Example values	Notes
Covidence #	integer	Covidence record identifier for the included study.		
Study ID	text	Study ID (First author year) for QUADAS-2 assessment.		
Title	text	Study title in the assessment sheet.		
SQ1	categorical	QUADAS-2 signalling question 1 response.	Low risk of bias; High risk of bias	
SQ2	categorical	QUADAS-2 signalling question 2 response.	Low risk of bias; High risk of bias	
SQ3	categorical	QUADAS-2 signalling question 3 response.	Low risk of bias; High risk of bias	
SQ4	categorical	QUADAS-2 signalling question 4 response.	Low risk of bias; High risk of bias	
Domain 1: Patient Selection/Study Design	categorical	Overall risk of bias judgement for QUADAS-2 Domain 1.	Low risk of bias; High risk of bias	
SQ5	categorical	QUADAS-2 signalling question 5 response.	Low risk of bias; High risk of bias	
SQ6	categorical	QUADAS-2 signalling question 6 response.	Low risk of bias; High risk of bias	
Domain 2: Index Measure	categorical	Overall risk of bias judgement for QUADAS-2 Domain 2.	Low risk of bias; High risk of bias	
SQ7	categorical	QUADAS-2 signalling question 7 response.	Low risk of bias; High risk of bias	
SQ8	categorical	QUADAS-2 signalling question 8 response.	Low risk of bias; High risk of bias	
Domain 3: Criterion Measure	categorical	Overall risk of bias judgement for QUADAS-2 Domain 3.	Low risk of bias; High risk of bias	
SQ9	categorical	QUADAS-2 signalling question 9 response.	Low risk of bias; High risk of bias	
SQ10	categorical	QUADAS-2 signalling question 10 response.	Low risk of bias; High risk of bias	
SQ11	categorical	QUADAS-2 signalling question 11 response.	Low risk of bias; High risk of bias	
Domain 4: Flow and Timing	categorical	Overall risk of bias judgement for QUADAS-2 Domain 4.	Low risk of bias; High risk of bias	

Study_characteristics.csv

Rows: 140. Columns: 35.

Column	Type	Description	Allowed/Example values	Notes
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_characteristics.id to validation-level tables.
study	text	Study ID in the format First author (year).		Corresponds to 'Study ID' in the extraction manual.
year	integer	Publication year of the study.		
country	text	Countries where the study was conducted (semicolon-separated if multiple).		
age_group	categorical	Age group category for the cohort used for model development/testing.	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
sample_size	integer	Total recruited sample size for the study (study-level).		
females	integer	Number of female participants (if reported).		If not reported, coded as n.r. at extraction time.
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries.
ethnicity_reported	categorical	Whether ethnicity distribution was reported.	Yes; No; n.r.	
unique_model	text	Indicator that the study contributed a unique prediction model.		
device	text	Wearable device model as reported (study-level).		
device_brand	text	Wearable device brand.		
sensor	text	Accelerometer sensor chip/model where reported.	ADXL330 (Analog Devices); ADXL335 (Analog Device); BMA150 (Bosch Sensortec); KXSC7-3672 (Kionix); n.r.	
sensor_type	categorical	Accelerometer sensor type.	Uniaxial; Biaxial; Triaxial; Other; n.r.	
dynami_range	text	Full-scale/dynamic range of accelerometer (e.g., ±8 g).	±6g; ±16g; ±5g; n.r.; ±1.5g; ±16g; ±2g; ±3g; ±4g; ±5g; ±6g; ±6g; ±8g; ±8g; ±8g; ±16g; ±8g	Column header spelling retained from source file.

bit_resolution	text	Bit resolution of the accelerometer output (e.g., 12-bit) or other reported resolution.	10 bit; 12 bit; 13 bit; 16 bit; 4 mg; 8 bit; n.r.	
acquisition_sampling_rate	text	Native sampling rate used for data collection (Hz).		
sampling_rate_cat	categorical	Categorised sampling rate for reporting.	1-20 Hz; 21-50 Hz; 51-80 Hz; 81-100 Hz; >100 Hz; Variable; n.r.	
resampled	categorical	Indicator that raw data were resampled prior to feature extraction.	Yes; No; n.r.	
resampled_sampling_rate	text	Target sampling rate after resampling (Hz), if applicable.		
raw_signal_check	categorical	Whether calibration/quality control checks on raw signals were reported.	Yes; No; n.r.	
digital_filter	categorical	Whether a digital filter was applied (as reported).	Yes; No; None; n.r.	
digital_filter_type	text	Filter type (e.g., low-pass, band-pass), if applied.	Band-pass; High-pass; Low-pass; n.r.	
highpass_cutoff	text	High-pass cutoff frequency (Hz), if applicable.		
lowpass_cutoff	text	Low-pass cutoff frequency (Hz), if applicable.		
gravity_removal	text	Approach used to treat/remove gravitational component.	De-trend subtraction; FIlter; Subtracting 1g from VM; None; Subtracting mean; Subtracting prior 5 sec moving average; n.a.	
feature_types	text	Feature domains extracted (study-level).		Semicolon-separated list.
features	text	Features extracted or used in the model (context-specific).		
features_extracted_from	text	Signal source used for feature extraction (axes, vector magnitude).		
window_size	text	Window length/epoch used for feature extraction.		
feature_types_model	text	Feature domains used as model inputs		

		after selection.		
feature_selection	text	Feature selection approach.		
method_best	text	Best-performing method (study-level summary).	Classical ML; Deep Learning; Heuristic Rule; Hybrid / Ensemble; Linear / GLM; Non-linear / Mixed Statistical; Probabilistic Temporal; Threshold / Cutpoint	
sub_method_best	text	Best-performing sub-method (study-level summary).		
outcomes	text	Movement behaviour outcomes assessed in the study (semicolon-separated).	Activity intensity; Activity type; Energy expenditure; Sleep-Wake	